MIL-C-1911J
9 February 1988
SUPERSEDING
MIL-C-1911H
6 March 1980

MILITARY SPECIFICATION

CAP, CAMOUFLAGE PATTERN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a cap made from cotton and nylon with flannel lined earlaps.

1.2 Classification. The cap shall be of one type in the following sizes (see 6.2):

Schedule of sizes

6-3/8	6-7/8	7-3/8
6-1/2	7	7-1/2
6-5/8	7-1/8	7-5/8
6-3/4	7-1/4	7-3/4

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

A-A-203	-	Paper, Kraft, Wrapping
UU-P-31	-	Paper; General Specifications and Methods of Testing
DDD-L-20	-	Label; for Clothing, Equipage and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-T-3530	-	Thread and Twine; Mildew-Resistant or Water-Repellent Treated
MIL-C-3760	-	Cloth, Flannel, Wool and Cotton; Wool and Nylon
MIL-C-3924	-	Cloth, Oxford, Cotton Warp and Nylon Filling, Quarpel Treated
MIL-C-11065	-	Cloth, Flannel, Wood and Nylon, Shrink Resistant
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-T-43548	-	Thread, Polyester Core: Cotton-, Rayon-, or Polyester-Covered

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

COLOR ASSOCIATION OF THE UNITED STATES, INC.

Department of Defense Standard Shades

(Application for copies should be addressed to the Color Association of the United States, Inc., 343 Lexington Avenue, New York, NY 10016.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case this document shall govern.

3.2 Materials (see 6.5).

3.2.1 Basic material. The basic material for the cap shall be cotton warp, nylon filling oxford (Quarpel treated) cloth, conforming to class 3, camouflage printed of MIL-C-3924.

3.2.1.1 Inside crown, inside crown side pieces and crown seam reinforcement strip. The cloth for the inside crown, inside crown side pieces and crown seam reinforcement strip shall be either class 3 or class 2 dyed OG-107 material of MIL-C-3924. As an alternative, the cloth may be printed seconds or dyed the ground shade prior to printing of class 3 material of MIL-C-3924 or the cloth may be dyed seconds of class 2 material of MIL-C-3924. Printing seconds shall be cloth which has been rejected only for defects cited in MIL-C-3924 pertaining to color infra-red reflectance or pattern of camouflage print. If printed seconds are used they must be marked on the face side with stripes 3 inches apart the length of the roll. The color of the stripes shall approximate any of the four camouflage print colors. Colorfastness to crocking shall be no lower than 6.5 (Munsell scale). Dyed seconds shall be cloth which has been rejected only for defects cited in MIL-C-3924 pertaining to color and the shade shall approximate OG-107.

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3.2.2 Lining. The lining for the ear flaps shall be wool and nylon flannel, OG-108 conforming to type III of MIL-C-3760 or to MIL-C-11065 or shall be a double face polyester fleece cloth as specified in 3.2.2.1.

3.2.2.1 Double face polyester fleece cloth. The cloth shall be a double face polyester fleece cloth in a circular knit construction made of 150 denier continuous filament polyester face and back yarns and shall match OG-108. The face pile shall be evenly sheared and napped before panne. The face of the cloth shall be marked "FACE". The cloth shall conform to the finished requirements shown below when tested as specified in 4.2.1.1.

Characteristic	Requirement
Weight, oz/yd^2	6.5 to 7.5
Finished thickness, inches (max) at 0.1 psi	0.06
Fabric construction (wales x courses)	20 x 28
Busting strength, pounds (min)	125
Shrinkage, percent (max) (wales x courses)	3.0 x 3.0
pH	4.5 to 8.5
Colorfastness to:	
Laundering (min)	Good
Crocking (min)	Good
Perspiration (min)	Good

3.2.3 Visor interlining. The material for interlining the visor shall be a non-woven, fibrous material produced from a highly purified alpha cellulose vegetable fiber, wet web impregnated with compound neoprene and conforming to the following physical requirements when tested as specified in 4.2.1.1.

	Machine direction	Cross direction
Stiffness (load lb. per in., min.):		
Original	2.4	1.5
After laundering (3 cycles)	2.4	1.5
After wet-drycleaning	2.2	1.4
Tensile strength (lb. per in., min.)	156	103
Stretch (percent, max.)	17	26
Tearing strength (grams, min.)	1770	2040
Weight - 2.21 lbs. per sq. yd. $\pm$ 5 percent		
Thickness - 0.075 $\pm$ 0.003 inch		

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3.2.3.1 Water absorption. The finished material shall not absorb more than 30 percent water, when tested as specified in 4.2.1.1.

3.2.3.2 Perspiration. The finished material shall show no embrittlement, development of stickiness, or delamination, when tested as specified in 4.2.1.1.

3.2.3.3 Accelerated aging. The finished material shall show no embrittlement, development of stickiness, or delamination, when tested as specified in 4.2.1.1.

3.2.3.4 Shrinkage. The finished material shall not shrink or warp, when tested as specified in 4.2.1.1.

3.2.3.5 Directional marking. The finished non-woven, fibrous material shall be marked by directional symbols to designate the greater flexibility of the material. The marking shall be legibly and clearly designated on the textured side of the material.

3.2.4 Thread. The thread shall be polyester cotton wrapped, dyed Olive Drab S-1, C.A. 66022, ticket Nos. 30, 50, and 70 conforming to MIL-T-43548. All thread shall be water repellent treated in accordance with type II, class 3 of MIL-T-3530. The dyed thread shall show fastness to light and laundering equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show "good" fastness to light and laundering.

3.2.5 Labels. Each cap shall have a combination identification - size label, and an instruction label, conforming to DDD-L-20. Label color shall be Medium Green, Cable No. 70034, 70130, or 70131 and shall show fastness to laundering in accordance with DDD-L-20 requirements. As an alternate, the labels may be combined into one label conforming to class 14.

3.2.5.1 Combination identification - size label. The combination identification - size label shall conform to type VI, class 4.

3.2.5.2 Instruction label. The instruction label shall conform to type VI, class 3, and shall contain the following information:

CAP, CAMOUFLAGE PATTERN

1. If cap is worn under helmet, helmet head band may require readjustment for proper fit and comfort.

2. Machine Wash. Use permanent press cycle. Wash in warm water with mild detergent.

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3. Hand Wash. Hand wash in warm water using mild detergent. Do not wring or twist. Rinse in clean warm water.

4. Do not use chlorine bleach or starch.

5. Dry at low heat (Do not exceed 130°F).

DO NOT REMOVE THIS LABEL

3.3 Design. The cap is of a visor style, having a circular top crown, two side crown pieces, outside crown band, hinged visor, and retractable earlap. The cap is lined with self material except for the flannel lined earlap (see figure 1.)

3.4 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government. The working patterns shall be identical to the standard patterns which shall not be altered in any way. The patterns provide a 3/8 inch seam allowance for seaming earlaps and visor to lining, joining side pieces for crown and side pieces for lining, and 1/4 inch for all other seams. Notches are provided on the side pieces to designate the top and on the crown to designate the front.

3.4.1 Pattern parts. The components of the cap shall be cut from the materials, as specified below, in accordance with the pattern parts indicated.

TABLE I. List of pattern parts

Material	Pattern nomenclature	Cut parts	
Cloth, oxford	Top visor cover	1	
	Under visor cover	1	
	Side pieces for crown	2	
	Side pieces for crown lining	2	
	Outside band	1	1/
	Earlap	1	
	Crown	1	
	Crown lining	1	
	Crown seam reinforcement strip	1	
Wool and nylon flannel or polyester double face fleece	Earlap lining	1	
Alpha-cellulose interlining fiber	Visor interlining	1	

1/ Band may be cut on a roll in sufficient width to meet the finished requirements of table II, operation 6.

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3.5 Construction. The construction shall conform in all respects to the requirements specified in table II and herein. Figure 1 is furnished solely for guidance and information. Should variation from the document appear in figure 1, the document shall govern.

3.5.1 Stitches, seams, and stitching. All stitches, seams, and stitching shall conform to FED-STD-751. The type of seam, stitching, and stitches per inch shall be as specified in table I. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers, or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used. Where stitch type 401 is used, the chain or underside of the stitch shall be on the inside of the cap. The gage of stitching for safety construction shall be 3/8 inch with the guides and knives set to trim only the ravelled ends of the fabric.

3.5.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/2 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall overlap not less than 1/2 inch. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewn.

3.5.1.1.1 Repairs of type 301 stitching.

a. When thread breaks, skipped stitches, run-offs, or bobbin runouts occur during sewing, except prestitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch in back of the end of the stitching. 1/

b. Except for prestitching, thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.5.1.2 Type 401, 502, 503 and 504 stitching. Thread tension shall be maintained so that there will be no loose or excessively tight stitching. All repairs shall be in accordance with 3.5.1.1.1a or b. Repairs to stitch type 401 may be accomplished by use of stitch type 301.

3.5.2 Repairs. Repairs such as mends, darns, patches, or splices are not permitted on the cap.

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3.5.3 Automatic stitching. Automatic machines may be used to perform any of the stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met; and at least three tying, overlapping, or backstitches are used to secure the ends of the stitching.

3.5.3.1 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

3.6 Manufacturing operations requirements. The cap shall be manufactured in accordance with operations requirements specified in table II. The contractor is not required to follow the exact sequence of operations listed. Any additional basting or holding stitching to facilitate manufacture is permissible provided the thread is removed or does not show on the finished cap.

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NO.	TABLE II. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut the caps in strict accordance with patterns furnished, which show size, directional lines (warp or filling direction), and notches for the proper assembly of all parts. b. Cut the oxford top crown and visor pieces in the warp direction. Cut the crown seam reinforcement strip on the bias 3/4 ± 1/16 inch wide. Cut all other oxford parts in the filling direction, so that all outside edge stitching will be parallel to the fabric filling threads. An an alternative, the outside band may be cut in the warp direction. c. Cut the flannel earlap lining on the warp or filling. d. Cut the visor interlining, in accordance with directional lines shown on pattern. e. Cut all outside component parts out of one piece of material, except the crown seam reinforcement strip and outside band (when band is cut on roll).						
2.	Replacement of damaged parts. a. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in 4.2.2 shall be removed from production and replaced with non-defective and properly matched components.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Component marking. Mark, ticket or bundle all parts, to insure uniform shade, size, and proper assembly throughout the cap. Any method of marking may be used except: a. Corrosive metal fastening devices. b. Sew-on tickets. c. Adhesive tickets which leave an impression or traces of paper or adhesive on the material after removal of the ticket.						
4.	Make visor. Finished appearance. The visor shall be smooth and free of wrinkles and puckers. The hinge construction shall be evenly formed. The interlining shall not be caught in the stitching which forms the hinge. The finished front turned edge on top of visor shall be 1/4 inch wide. The quilting of the visor shall be uniform.						
	a. Seam top and under plies of visor together 3/16 to 1/4 inch from edge, distributing fullness evenly.	301 or 401	SSa-1 SSa-1	10-14 10-14	50 50	50 70	
	b. Place seamed plies on top of interlining, edge to edge, with the top ply next to interlining and stitch through plies and interlining 3/16 to 1/4 inch from outer edge.	301 or 401	SSa-1 SSa-1	6-8 6-8	50 50	50 70	
	c. Turn underply over interlining and raise stitch through all plies not more than 1/8 inch outward from turned seam.	301		6-8	50	50	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Make visor. (cont'd)						
	d. Quilt visor with six rows of parallel stitching spaced 3/16 to 1/4 inch apart, through all plies, starting 3/16 to 1/4 inch from the row of raised stitching.	301	SSv-6	6-8	30 or 50	30 or 50	
	e. Seam visor cover together along inner edge of interlining with two rows of stitching (not through interlining). The first row shall be not more than 1/16 inch from interlining with the second row 1/4 ± 1/16 inch from the first row.	301	SSa-2	6-8	30	30	
5.	Make side pieces.						
	Finished appearance. The side pieces shall be neatly seamed, without puckers or pleats and with the raise stitching evenly gaged.						
	a. Safety stitch front ends of side pieces; turn and raise stitch 1/16 to 1/8 inch from seam.	515 or 516 or 519	LSbm-3	10-14	50	70	
6.	Make band and body.						
	Finished appearance. The band shall be neatly and smoothly seamed to the side body, with top ends of band evenly matched in closing back center seam, and finish uniformly 1-1/8 ± 1/8 inches in width. The crown shall be neatly seamed, without puckers or pleats, and the raised stitching evenly gaged.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Make band and body. (cont'd)						
	a. Turn in the upper edge of band 1/4 to 5/16 inch and seam to the side pieces, not more than 1/8 inch below the turned edge, with the bottom edge of the band and side pieces even.	301 or 401	LSd-1 LSd-1	10-14 10-14	50 50	50 70	
	b. Seam bottom edge of band to side pieces, not more than 1/8 inch from edge.	301 or 401	SSa-1 SSa-1	10-14 10-14	50 50	50 70	
	c. Safety stitch back ends of side pieces with band; turn and raise stitch 1/16 to 1/8 inch from seam.	515 or 516 or 519	LSbm-3	10-14	50	70	
	d. Turn up bottom joined edges to form a 1/4 inch hem and seam 1/16 to 1/8 inch from folded edge. or Form the bottom 1/4 inch hem after operation 11.	301 or 401	EFa-1	10-14	50	50	
	e. Safety stitch top of crown to side pieces, catching the 3/4 inch bias oxford strip in the seaming; turn and raise stitch 1/16 to 1/8 inch from seam, with side pieces lapping crown.	515 or 516 or 519	LSbm-3	10-14	50	70	
7.	Make lining. Finished appearance. The lining pieces shall be neatly joined without twists, puckers, or pleats and with joining seams matching the outer crown pieces. The top and side pieces shall be well shaped.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Make lining. (cont'd)						
	a. Safety stitch front ends of side pieces; turn and raise stitch 1/16 to 1/8 inch from seam.	515 or 516 or 519	LSbm-3	10-14	50	70	
	b. Safety stitch back ends of side pieces; turn and raise stitch 1/16 to 1/8 inch.	515 or 516 or 519	LSbm-3	10-14	50	70	
	c. Safety stitch crown lining pieces to side pieces turn and raise stitch 1/16 to 1/8 inch from seam, with the side pieces lapping the crown.	515 or 516 or	LSbm-3	10-14	50	70	
8.	Labeling.						
	a. Center identification-size label on crown lining. 1 ± 1/2 inch forward from crown seam at back of cap. Stitch on four sides 1/16 to 1/8 inch from edge.	301 or 401		10-14 10-14	50 50	50 70	
	b. Position and center instruction label directly below identification-size label and stitch on all four sides not more than 3/16 inch from edges of label. Top of instruction label may be caught in the bottom row of stitches of identification label. or	301 or 401		10-14 10-14	50 50	50 70	
	c. Center combination identification, size, and instruction label on crown lining, 1 ± 1/2 inches forward from crown seam at back of cap and stitch on all four sides 1/16 to 1/8 inch from edge.	301		10-14	50	50	

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					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Make earlap. Finished appearance. The earlap shall finish without pleats or puckers. Overedge of flannel shall be neat without any raw edges. The bottom and front curved edge shall be joined so that earlap is slightly cupped to hug neck and ears.						
	a. Overedge earlap lining at top edge.	504 or 503 or 502	EFd-1 EFd-1 EFd-1	6-10 6-10 6-10	70 70 70	70 70 70	
	b. Position earlap liner to earlap as indicated by marks on pattern and stitch.	301 or 401	SSe-2(a) SSe-2(a)	10-14 10-14	50 50	50 70	
	c. Turn and single stitch 1/8 to 3/16 inch from edge continuing the stitching to ends of earlap.	301	SSe-2(b)	50	50		
	d. Seam top edge of flannel lining to earlap 1/16 to 1/8 inch from edge.	301	SSa-1	10-14	50	50	
10.	Set visor and earlap to crown lining. Finished appearance. The visor shall finish with a 1/8 inch wide hinge between cap and visor interlining. The earlap shall finish with the top edge of flannel lining 1/8 to 1/4 inch from bottom of cap at center back and 1/2 to 3/4 inch at front ends, with the ends of earlap lapping on the ends of the under side of the visor.						

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Set visor and earlap to crown lining. (cont'd)						
	a. Center earlap on back crown lining and center visor on front crown lining (overlapping ends of earlap, to back and front seams, respectively).						
	b. Seam bottom crown lining to earlap and visor 3/8 inches from edge, automatically forming a 1/8 ± 1/16 inch hinge on the visor.	301 or 401	LSq-2(a) LSq-2(a)	10-14 10-14	30 30	30 50	
	c. Turn crown lining up and raise stitch 1/16 to 1/8 inch from edge.	301	LSq-2(b)	10-14	30	30	
11.	Assemble cap.						
	Finished appearance. The outer shell and lining shall be evenly joined without fullness, pleats, or puckers.						
	a. Position lining in outer shell, matching front and back seams plus or minus 1/4 inch, maintaining visor hinge and seam around 1/16 to 1/8 inch from bottom edge of outer shell.	301	LSe-1 shows finished seams	10-14	50	50	
12.	Clean caps.						
	a. Trim all ends of thread to 1/4 inch maximum length.						
	b. Remove all loose threads, spots, stains, and shade and size tickets. Remove pin tickets without damage to the material.						

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3.7 Measurements. The finished caps shall conform to measurements for each respective size, as shown in table III. All caps shall be measured with earlap inside the cap.

TABLE III. Cap measurements (in inches)

Size	Inside cap circumference (± 1/4 inch tolerance)
6-3/8	20-1/8
6-1/2	20-1/2
6-5/8	20-7/8
6-3/4	21-1/4
6-7/8	21-5/8
7	22
7-1/8	22-3/8
7-1/4	22-3/4
7-3/8	23-1/8
7-1/2	23-1/2
7-5/8	23-7/8
7-3/4	24-1/4

3.8 Workmanship. The finished caps shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicted or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.1.1 Component testing. In addition to testing provisions contained in subsidiary specifications or standards, testing shall be performed in accordance with table IV for characteristics noted. When data in the "Determinations per sample unit" and "Results reported as" columns are not specified in table IV, they shall be reported as required by the referenced test method. All test reports shall contain the individual values used in expressing the final result. The lot shall be unacceptable if one or more sample units fail to meet any of the test requirements specified. The lot size shall be expressed in units of yards. The sample unit shall be 1 square yard. The sample size shall be in accordance with the following:

Lot size	Sample size
800 or less	2
801 thru 22,000	3
22,001 and over	5

TABLE IV. Component tests

Component	Characteristic	Requirement paragraph	Test method	Determinations per sample unit	Results reported as
Polyester double face fleece lining	Yarn denier	3.2.2.1	1/	---	
	Knit	3.2.2.1	Visual	---	---
	Color	3.2.2.1	Visual	---	---
	Weight	3.2.2.1	5041 2/	---	Nearest 0.1 oz/yd^2
	Thickness	3.2.2.1	5030 2/	---	Nearest 0.01 inch
	Wales and courses per inch	3.2.2.1	5070 2/	---	Nearest whole number
	Bursting strength	3.2.2.1	5120 2/	---	Nearest 1 pound
	Shrinkage	3.2.2.1	AATCC 135-1978 6/	---	Nearest 0.1 percent

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TABLE IV. Component tests (cont'd)

Component	Characteristic	Requirement paragraph	Test method	Determinations per sample unit	Results reported as
Polyester double face fleece lining (cont'd)	pH	3.2.2.1	2811 2/	---	Nearest 0.1
	Colorfastness to:				
	Laundering (After 3 cycles)	3.2.2.1	5610 7/	---	Pass or fail
	Crocking	3.2.2.1	5651	---	Pass or fail
	Perspiration	3.2.2.1	5680	---	Pass or fail
Non-woven fibrous material (for visor)	Impregnation compound	3.2.3	1/	---	---
	Stiffness:				
	Original	3.2.3	5202 2/	---	Nearest 0.1 lb. per inch
	After laundering (3 cycles)	3.2.3	5202 and 5610 2/	---	---
	After wet-dry-cleaning	3.2.3	5202 and 5622 2/	---	---
	Tensile strength	3.2.3	171 3/	---	Nearest 1 lb. per inch
	Stretch elongation	3.2.3	171 3/	10 each direction	Nearest 1 percent
	Tearing strength	3.2.3	170 3/	---	Nearest 1 gram
	Weight	3.2.3	5041 2/	---	Nearest 0.01 lbs.
	Thickness	3.2.3	5030 2/	---	Nearest 0.001 inch
	Water absorption	3.2.3.1	5502 2/	---	Nearest 1 percent
	Perspiration	3.2.3.2	5680 2/	---	Pass or fail
	Accelerated aging	3.2.3.3	4/	1	Pass or fail
	Shrinkage	3.2.3.4	5/	1	Pass or fail

1/ A certification of compliance shall be submitted and will be accepted for the stated requirement.

2/ Refers to FED-STD-191.

3/ A 3 by 3 inch specimen shall be aged in a circulating air oven maintained at 100 ± 2°C for 15 days. After removal of the specimen from the oven, the specimen shall be conditioned and examined for embrittlement, stickiness, and delamination.

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4/ A 3 by 3 inch specimen shall be soaked in distilled water at 23 ± 2°C for 24 hours. After removal of the specimen from the water, the specimen shall be examined for warping and measured for shrinkage. Four measurements shall be taken across the width of the specimen (two in machine direction and two cross direction), 1 inch from each corner using a ruler graduated in 64's of an inch.

5/ Machine cycle 3, wash temp II, drying procedure B, except not to exceed 130°F.

6/ The specimen shall be dried after each cycle.

7/ Refers to UU-P-31.

4.2.2 End item visual examination. The end item shall be examined for the defects listed in table V. The lot size shall be expressed in units of one cap. The sample unit shall be one cap. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects, 10.0 for major and minor A defects combined, and 25.0 for total (major, minor A, and minor B) defects combined.

TABLE V. End item visual defects

Examine	Defects	Classification Major	Minor A	Minor B
Material defects and damages	a. Hole, cut, tear, needle chew or burn.	X		
	b. Drill hole on outside of cap.			X
	c. Visible mend.	X		
	d. Weakening defects:			
	1. Smash.	X		
	2. Multiple float or loose slub:			
	- on outside.	X		
	- on inside.		X	
	e. Permanent crease in cloth (when caught in seaming or stitching) on outside.	X		
	f. Unsightly slub or shade bar:			
	- on outside.		X	
	- on inside.			X
Shaded parts	Shade varying within a part or between two or more parts on outside except for outside band (when band is cut on a roll).		X	

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TABLE V. End item visual defects (cont'd)

Examine	Defects	Classification Major	Minor A	Minor B
Cleanness	a. Spot or stain, clearly noticeable:			
	- on outside.		X	
	- on inside.			X
	b. One or more marking tickets not removed.			X
	c. Any shade or size marking visible on outside.		X	
	d. Thread ends not trimmed, or loose threads not removed.			X
	e. Visible basting thread not removed.			X
Cutting	Any component part not cut in conformance with directional lines indicated on patterns, or not cut in accordance with specification requirements.	X		
Components and assembly	a. Any component part or required operation omitted, (unless otherwise classified herein).	X		
	b. Any component not as specified.	X		
	c. Any operation not performed as specified (unless otherwise classified herein).			X
	d. Crown reinforcement strip omitted.			X
Seams and stitching	a. Accuracy of seaming:			
	1. Seams or component part twisted, puckered or pleated (unless otherwise classified herein).		X	
	2. Part of material caught in unrelated operation or stitching.	X		
	3. Stitching overlapping end(s) of thread break less than 1/2 inch.			X
	4. Ends of stitching not secured as specified.			X
	b. Gage of stitching and seam allowances:			
	1. Irregular, i.e., unevenly gaged or various stitchings not uniform (to be scored only when condition exists on major portion of seam).	X		

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TABLE V. End item visual defects (cont'd)

Examine	Defects	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)	2. Beyond range of width specified or varies more than 1/16 inch when no range is specified.		X	
	c. Stitches, broken or missing (including overedging):			
	- up to and including 1/4 inch.			X
	- more than 1/4 inch but not more than 1/2 inch.		X	
	- more than 1/2 inch.	X		
	d. Skipped stitches or runoff stitches:			
	- 1/4 inch up to and including 1/2 inch long.			X
	- more than 1/2 inch long.		X	
	e. Raw edge more than 1/4 inch long.			X
	f. Seam or stitch type not as specified.			X
	g. Stitches per inch (to be scored only when condition exists on major portion of seam) less than minimum or more than maximum specified.			X
	h. Stitch tension:			
	1. Loose tension resulting in a loose seam.	X		
	2. Loose tension (exposed loops of top or bottom threads) more than 1/2 inch in length.			X
	3. Tight tension (stitches break when strain is applied in the direction of the seam or stitching).		X	
Labels	a. Identification and size label omitted, incorrect, or illegible.	X		
	b. Instruction label omitted, incorrect, or illegible.		X	
	c. Label not stitched on all four sides.			X
	d. Label not positioned as specified.			X

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TABLE V. End item visual defects (cont'd)

Examine	Defects	Classification Major	Minor A	Minor B
Band	a. Fullness in width, affecting appearance.		X	
	b. Less than 1 inch or more than 1-1/4 inches wide.		X	
Earlap	a. Off center of back seam:			
	- more than 1/2 inch.		X	
	- more than 1/4 inch to 1/2 inch inclusive.			X
	b. Bottom of front curved edge stretched, or flannel too full, causing edges to curl out preventing earlap from being slightly cupped to hug neck and ears.		X	
	c. Badly twisted.		X	
	d. Top edge of flannel from lining joining seam:			
	- less than 1/8 inch at center back.		X	
	- more than 1/4 inch at center back.			X
	- less than 1/2 inch at ends.		X	
	- more than 3/4 inch at ends.			X
Lining	a. Tight, short, or twisted causing fullness or twist on outside of cap.		X	
	b. Back or front seams of side pieces not seam on seam by more than 1/4 inch.		X	
Visor	a. Off center:			
	- 1/8 to 1/4 inch inclusive.			X
	- over 1/4 inch.		X	
	b. Quilted with less than six rows of stitching		X	
	c. One or both rows at back edge omitted.			X
	d. Interlining caught in visor to cap joining seam.	X		
	e. Poorly shaped, twisted, or full, affecting appearance.		X	

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4.2.3 End item dimensional examination. The end items shall be examined for the dimensions specified in 3.7. Any measurement not within the specified tolerance shall be classified a defect. The lot size shall be expressed in units of caps. The sample unit shall be one cap. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and unit pack)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flap, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container.
Content	Number of bundles per shipping container is more or less than required. Number of caps per bundle is more or less than required. 1/ Size on one or more caps not as specified on shipping container. 1/

1/ For this defect, two bundles shall be examined from each shipping container in the sample.

4.2.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded as specified.

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Examine	Defect
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Ten caps of one size only shall be nested together in a bundle with visors lying flat and cross-tied with cotton tape or twine.

5.1.2 Commercial. The caps shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Two hundred caps of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. In addition, each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.4. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except the type II, class F loads.

5.2.2 Level B packing. Two hundred caps of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approx-imate dimensions are furnished as a guide only. In addition, each shipping container shall have the contents completely covered on the top and bottom with a sheet of 30-pound

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minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except the inspection shall be in accordance with 4.2.4. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification, except that the inspection shall be in accordance with 4.2.4.

5.2.3 Commercial packing. The caps preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), caps packed as specified in 5.2.2 or 5.2.3 shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV or MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED NSN's" plainly printed or stamped thereon and under these words shall be legibly printed or stamped the correct quantity and NSN's contained therein.

6. NOTES

6.1 Intended use. The cap is intended for wear by military personnel in a temperate zone environment.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Size required (see 1.2).
c. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
d. Type and class of unit load required (see 5.2.1).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.1.2.1).
f. When palletization is required (see 5.3).

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6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document.

* 6.5 Subject term (key word) listing.

Camouflage
Cap
Clothing

6.6 Changes from previous issue. The margins of this document are marked with an asterisk to indicated where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies to these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-1911J	Cap. Camouflage Pattern

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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